

CHUR, Switzerland

WMO: 067860

Lat: 46.8703N

Lon: 9.5306E

Elev: 556

StdP: 94.82

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.6	(d) -13.7	(e) 1.2	(f) -6.2	(g) -11.5	(h) 1.5	(i) -4.9	(j) 9.1	(k) 4.6	(l) 8.2	(m) 5.0	(n) 1.7	(o) 200	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.1	19.2	29.0	18.6	27.1	18.0	20.1	28.6	19.3	27.1	18.6	25.6	4.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	13.3	22.4	16.7	12.7	21.8	16.0	12.2	21.4	60.1	28.6	57.3	27.5	54.9	25.6	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 7.3	(c) 6.5	(e) -11.2	(f) 34.0	(g) 2.7	(h) 1.7	(i) -13.2	(j) 35.2	(k) -14.7	(l) 36.3	(m) -16.3	(n) 37.2	(o) -18.2	(p) 38.5		
(a) 8.1	(b) 7.3	(c) 6.5	(e) -12.0	(f) 21.6	(g) 2.6	(h) 0.6	(i) -13.9	(j) 22.0	(k) -15.4	(l) 22.4	(m) -16.9	(n) 22.7	(o) -18.8	(p) 23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	1.4	2.8	6.9	10.6	14.8	18.2	19.8	19.4	15.2	11.6	6.1	2.4
	DBStd	7.56	3.97	4.43	4.10	3.94	3.70	3.96	3.48	3.41	3.45	3.78	4.25	3.88
	HDD10.0	1023	266	202	112	41	5	1	0	0	3	27	130	238
	HDD18.3	2944	524	434	356	231	121	51	25	28	104	208	367	496
	CDD10.0	1321	1	1	14	59	153	248	303	291	158	78	13	1
	CDD18.3	199	0	0	0	1	11	48	69	60	10	1	0	0
	CDH23.3	1646	0	0	0	11	112	427	572	456	62	6	0	0
	CDH26.7	504	0	0	0	0	21	133	192	152	7	0	0	0
(14) Wind	WSAvg	2.6	2.3	2.6	2.9	3.2	2.9	2.8	2.7	2.5	2.3	2.3	2.5	2.4
(15) Precipitation	PrecAvg	1094	64	67	68	69	93	123	130	141	97	79	90	73
	PrecMax	1517	167	229	177	151	162	236	217	240	189	184	289	199
	PrecMin	825	4	14	13	15	27	47	59	23	30	1	0	5
	PrecStd	158	43	51	38	32	41	48	38	46	44	43	57	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.7	20.5	24.9	29.1	32.7	33.5	33.6	27.3	23.9	18.8	14.1
		MCWB	6.7	7.4	10.6	13.4	16.4	19.3	19.6	19.5	17.6	14.5	11.2	8.0
	2%	DB	10.4	13.0	18.0	22.4	26.3	30.2	31.0	30.4	25.2	21.3	16.3	11.4
		MCWB	5.5	6.5	9.4	12.3	15.6	18.7	19.3	19.4	17.2	13.7	9.9	6.5
	5%	DB	8.5	11.0	15.7	20.2	24.1	28.0	28.8	28.0	23.1	19.4	14.2	9.3
		MCWB	4.5	5.3	8.4	11.4	14.5	18.0	18.8	19.1	16.2	12.9	9.0	5.5
	10%	DB	6.6	8.7	13.6	18.0	21.8	25.7	26.6	25.7	21.2	17.4	12.2	7.5
		MCWB	3.5	4.4	7.5	10.2	13.8	17.4	18.2	18.2	15.3	12.2	8.1	4.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.2	11.2	14.2	17.9	20.8	21.2	21.2	19.0	15.8	11.7	8.5
		MCDB	11.4	14.0	18.5	23.5	26.8	30.0	30.8	29.6	25.3	20.7	17.0	12.9
	2%	WB	6.1	6.8	9.9	12.9	16.5	19.5	20.1	20.1	17.8	14.6	10.4	7.2
		MCDB	9.4	11.9	16.6	21.2	24.3	28.0	28.7	28.2	23.6	19.3	15.0	10.6
	5%	WB	4.9	5.7	8.9	11.7	15.2	18.6	19.3	19.3	16.7	13.6	9.4	6.0
		MCDB	7.7	10.1	14.7	18.9	22.1	26.4	27.1	26.4	22.0	18.0	13.4	8.6
	10%	WB	3.9	4.6	7.9	10.7	14.2	17.7	18.5	18.5	15.8	12.7	8.4	4.9
		MCDB	6.2	8.2	12.6	17.0	20.5	24.6	25.5	24.7	20.0	16.7	11.8	7.2
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	9.4	10.3	10.2	10.3	10.3	9.7	9.1	8.8	7.1	6.2
		MCDBR	8.5	11.0	13.2	14.0	14.0	14.0	14.0	13.4	11.8	11.5	9.8	8.6
		MCWBR	5.1	6.1	6.8	6.5	5.9	5.7	5.3	5.5	5.8	5.7	5.1	4.9
		MCDBR	7.4	10.1	12.0	12.9	12.3	12.8	12.4	12.1	10.8	10.0	8.9	7.3
	5% WB	MCWBR	4.7	5.9	6.4	6.2	5.6	5.5	5.2	5.3	5.4	5.3	4.9	4.5
(40) Clear-Sky Solar Irradiance	taub		0.266	0.293	0.339	0.388	0.407	0.416	0.415	0.406	0.371	0.339	0.300	0.265
	taud		2.331	2.262	2.215	2.205	2.250	2.279	2.284	2.325	2.401	2.464	2.464	2.384
	Ebn at Noon		831	874	878	864	859	851	846	838	838	816	788	792
	Edh at Noon		73	97	119	133	132	129	127	117	99	80	65	62
(44) All-Sky Solar Radiation	RadAvg		1.44	2.36	3.57	4.42	4.85	5.65	5.58	4.85	3.73	2.43	1.45	1.20
	RadStd		0.17	0.29	0.50	0.68	0.49	0.51	0.49	0.46	0.37	0.28	0.20	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.38	N/A	N/A	+1.16	+0.47	+0.36	N/A	N/A	+87	+46
(47) Regional (116 neighbors)	N/A	N/A	N/A	+0.68	+0.40	+0.33	N/A	N/A	+59	+16

Nomenclature: See separate page